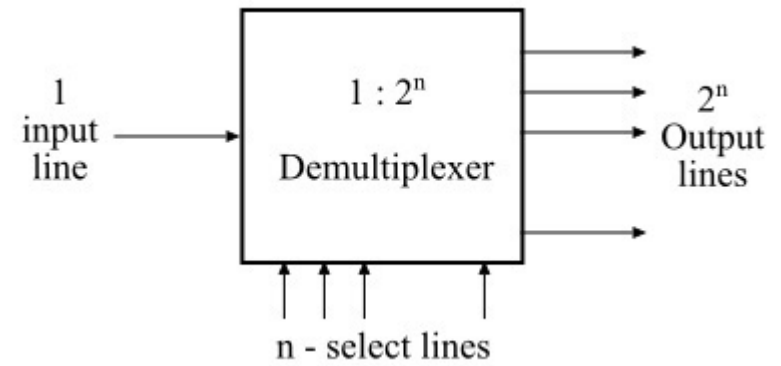
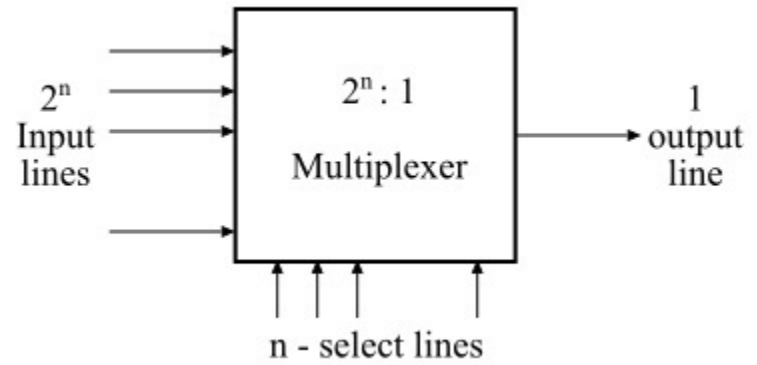
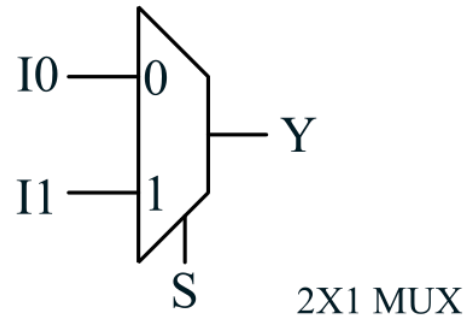
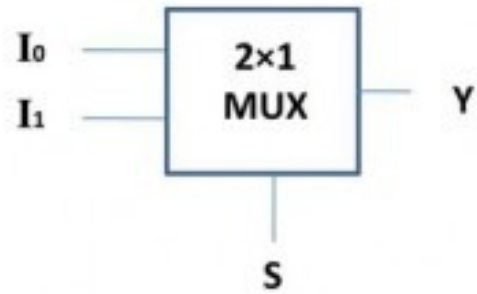


Multiplexer and De-multiplexer



2:1 Mux



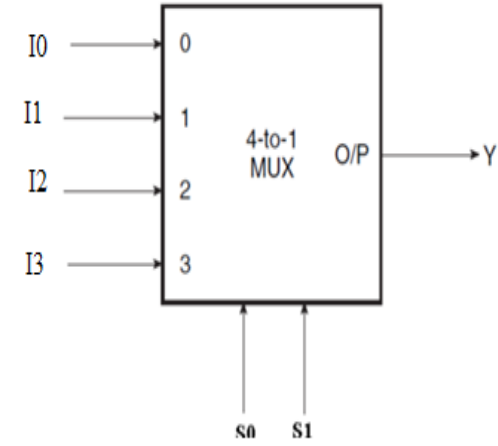
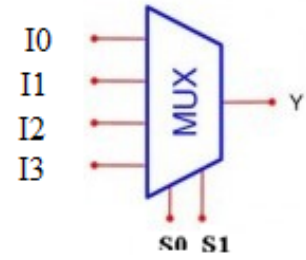
S	Y
0	I_0
1	I_1

Condensed Truth Table

S	I_0	I_1	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

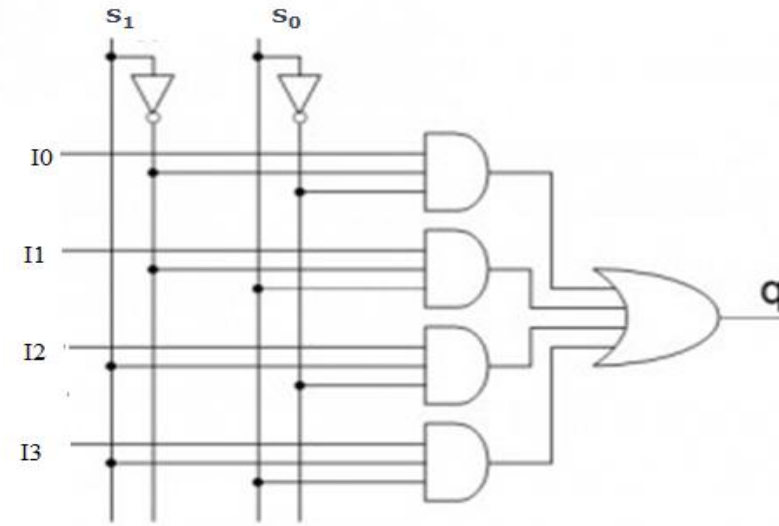
Truth table for 2x1 MUX

4:1 Mux



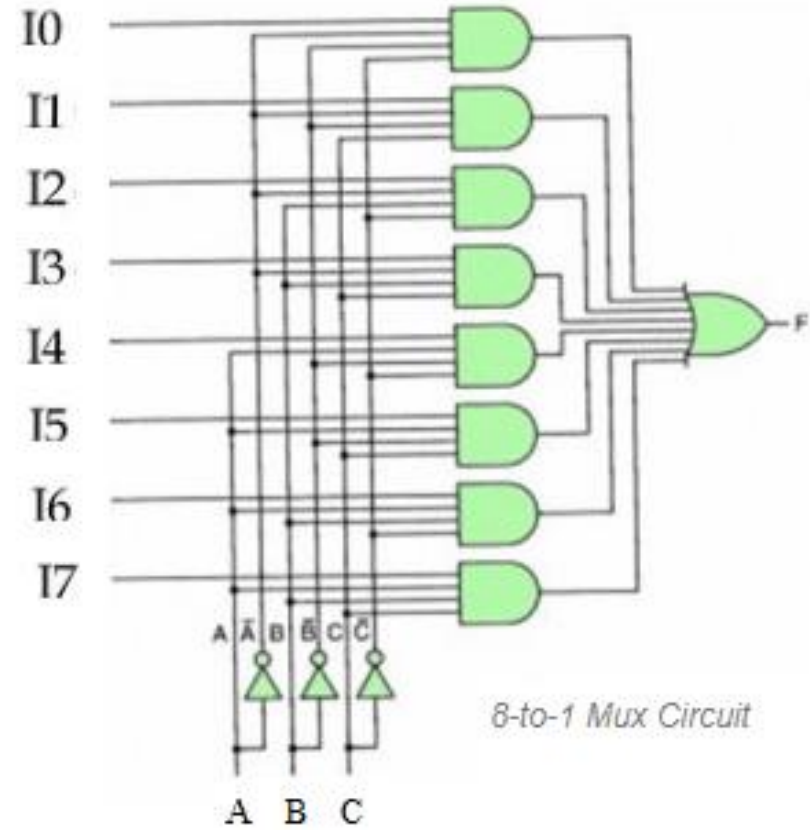
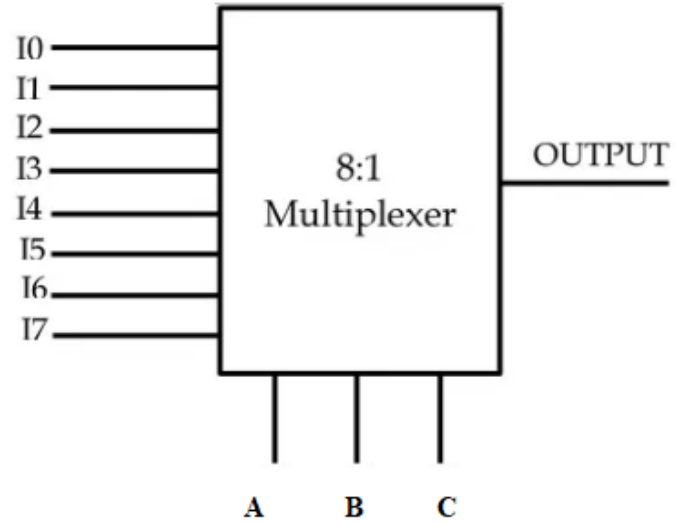
Selection Lines		Output
S ₁	S ₀	Y
0	0	I ₀
0	1	I ₁
1	0	I ₂
1	1	I ₃

$$Y = S_1'S_0'I_0 + S_1'S_0I_1 + S_1S_0'I_2 + S_1S_0I_3$$



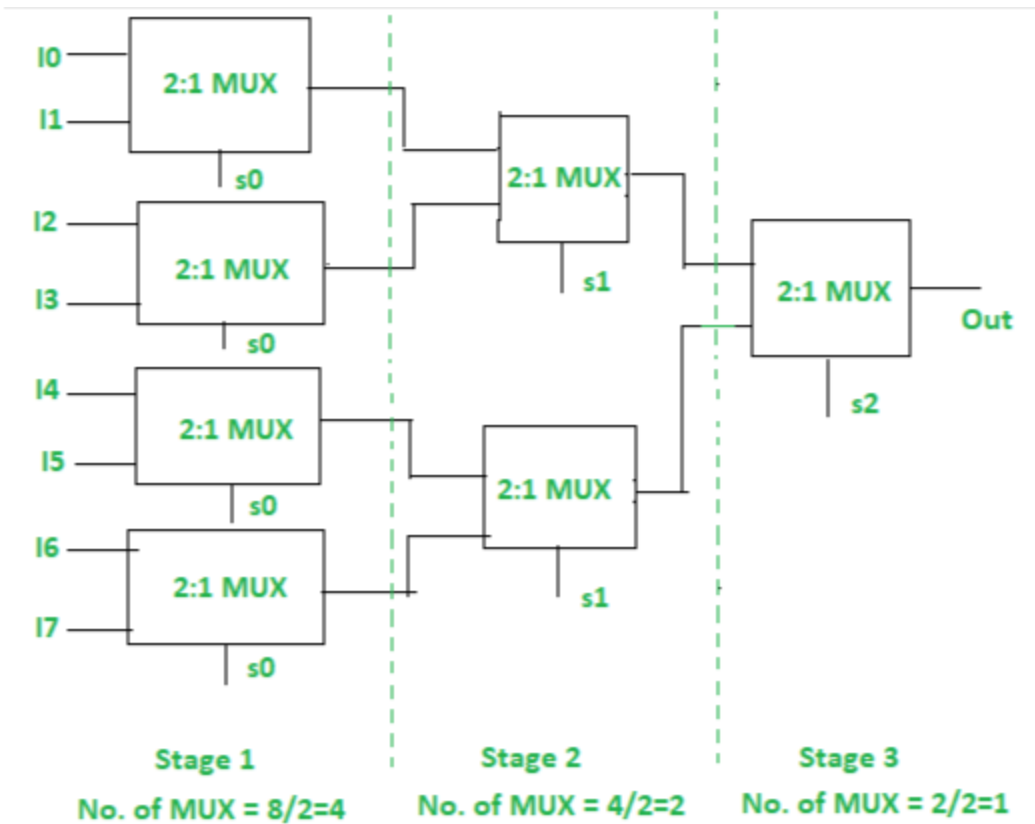
4X1 Mux

8:1 Mux



Mux Tree: (Design of higher order MUX with the help of lower)

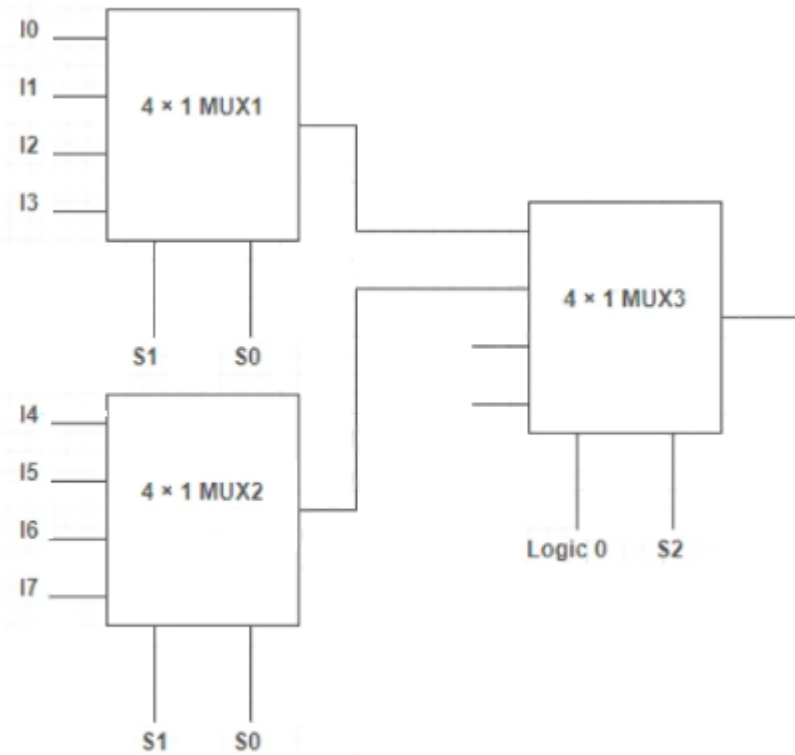
8:1 MUX using 2:1 MUX



S_2	S_1	S_0	Y
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

Mux Tree: (Design of higher order MUX with the help of lower)

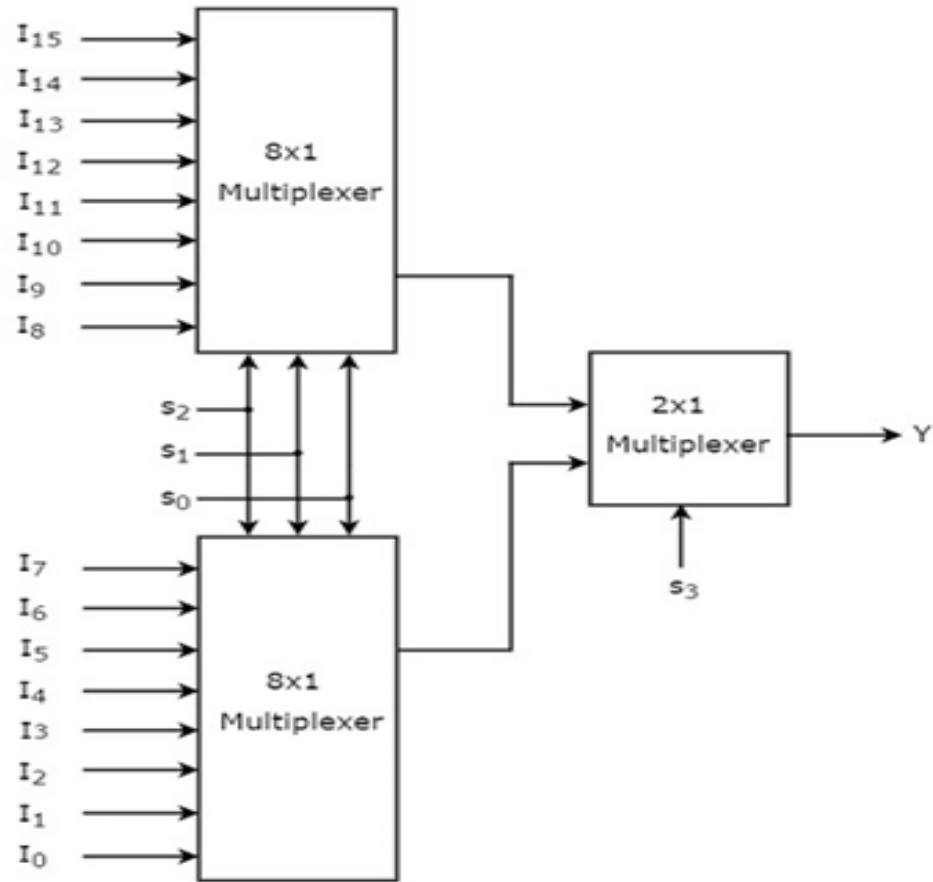
8:1 MUX using 4:1 MUX



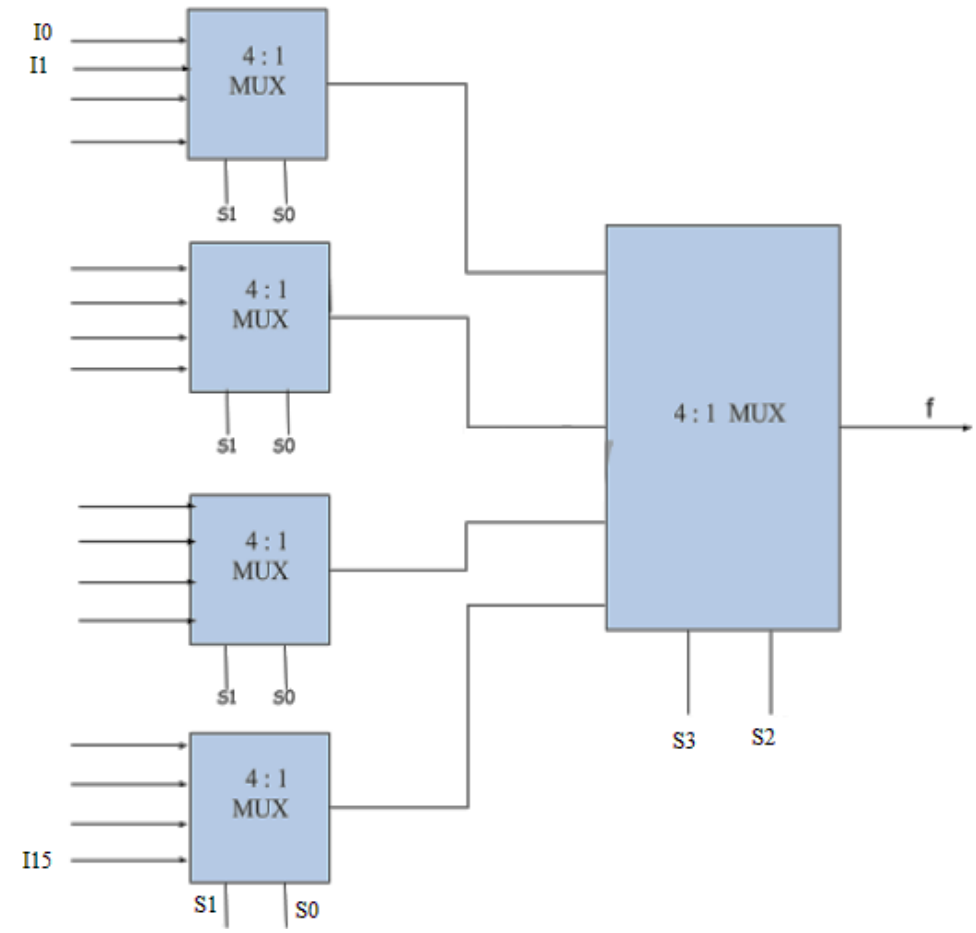
S_2	S_1	S_0	Y
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

Mux Tree: Design of higher order MUX with the help of lower

16:1 MUX using 8:1 MUX and 2:1 MUX



16:1 MUX using only 4:1 MUX

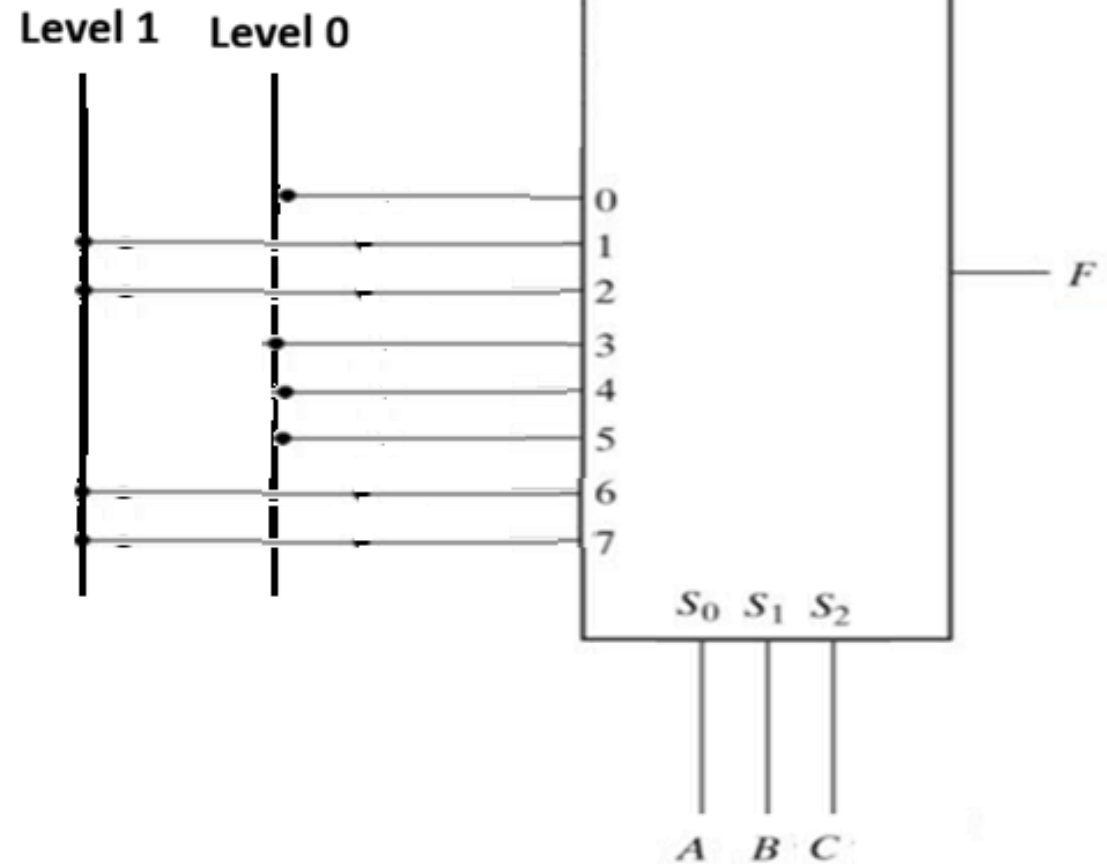


Realization of Boolean expression using 8:1 Mux

$$F = m_1 + m_2 + m_6 + m_7$$

<i>A</i>	<i>B</i>	<i>C</i>	<i>F</i>	
0	0	0	0	I0
0	0	1	1	I1
0	1	0	1	I2
0	1	1	0	I3
1	0	0	0	I4
1	0	1	0	I5
1	1	0	1	I6
1	1	1	1	I7

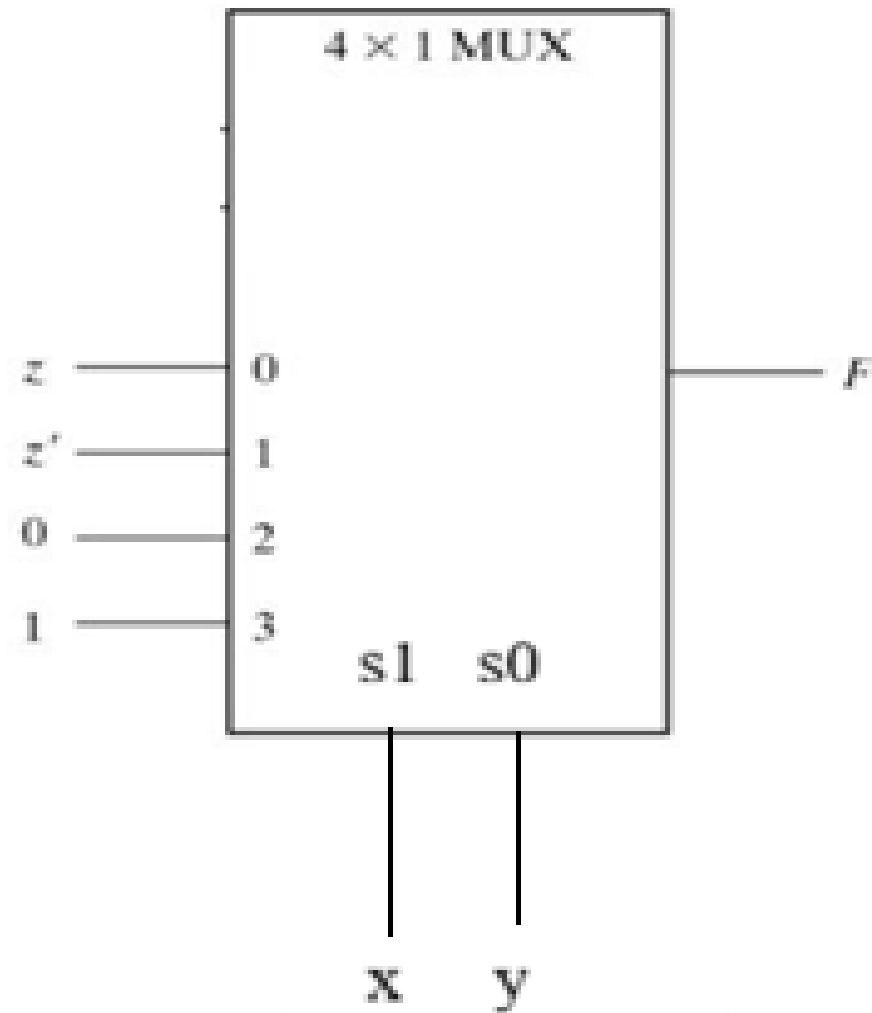
(a) Truth table



Realization of Boolean expression using 4: 1 Mux

x	y	z	F	
0	0	0	0	
0	0	1	1	$F = z$
0	1	0	1	
0	1	1	0	$F = z'$
1	0	0	0	
1	0	1	0	$F = 0$
1	1	0	1	
1	1	1	1	$F = 1$

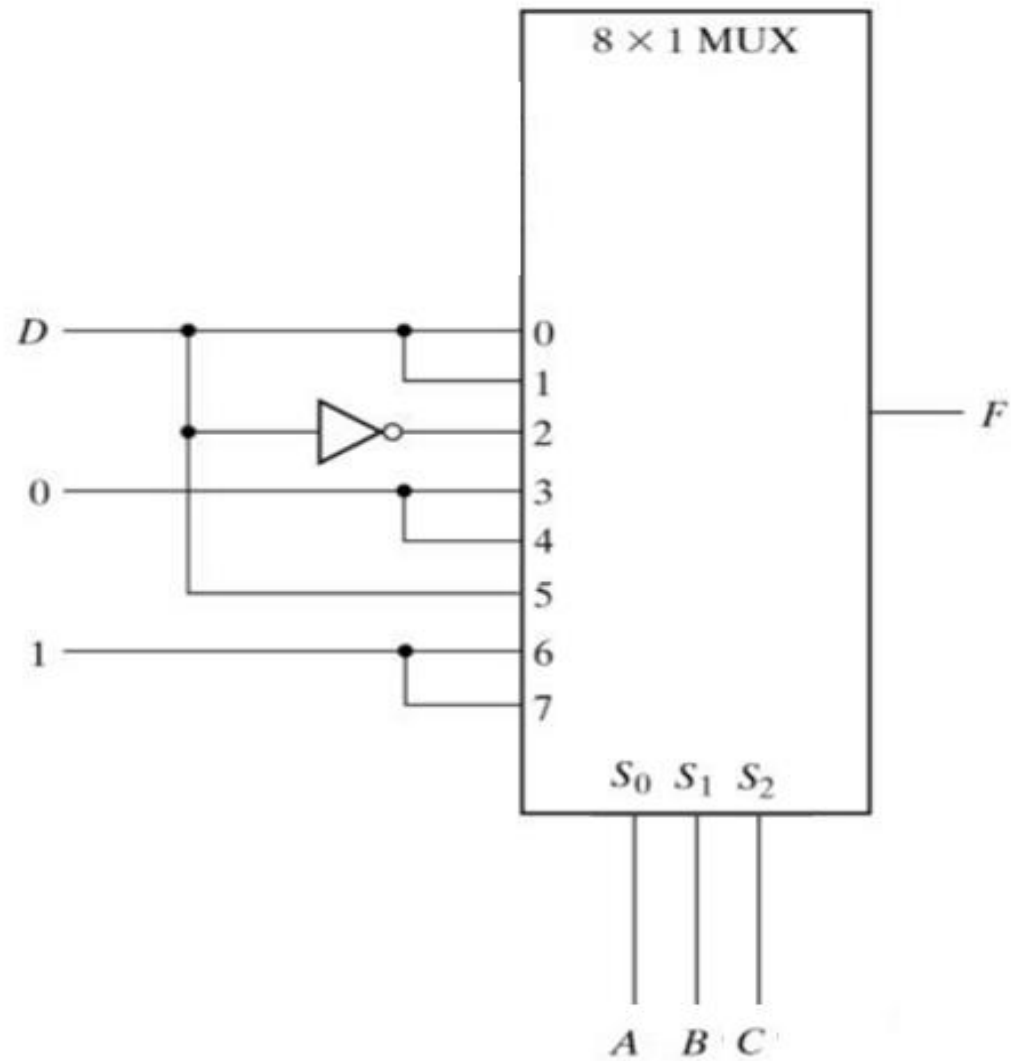
(a) Truth table



(b) Multiplexer implementation

Realization of Boolean expression using 8: 1 Mux

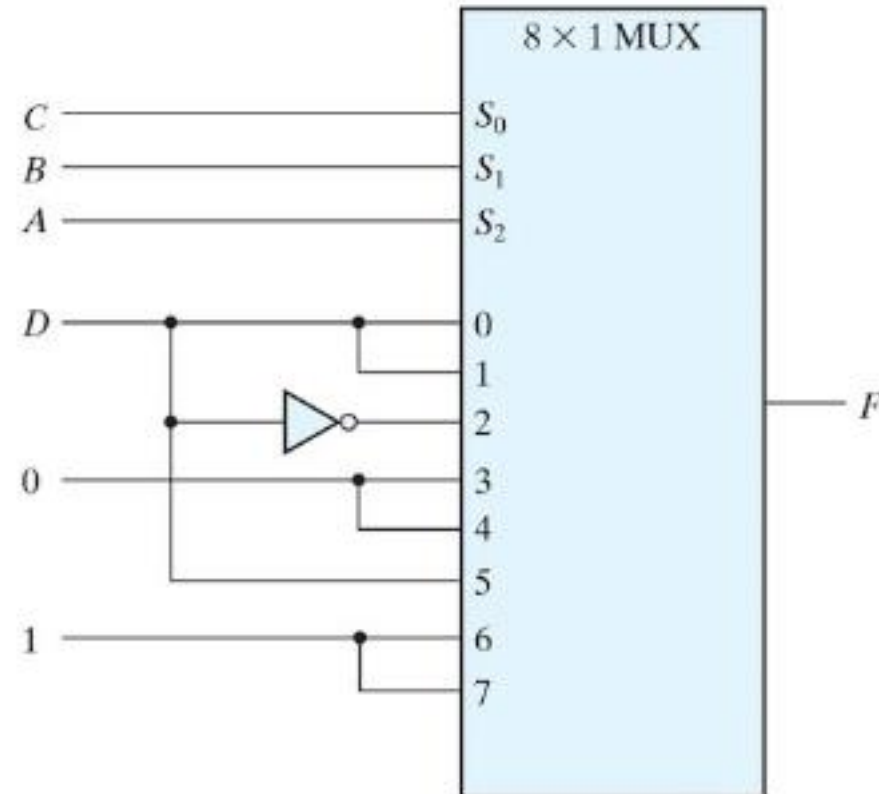
<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>F</i>	
0	0	0	0	0	
0	0	0	1	1	$F = D$
0	0	1	0	0	
0	0	1	1	1	$F = D$
0	1	0	0	1	
0	1	0	1	0	$F = D'$
0	1	1	0	0	
0	1	1	1	0	$F = 0$
1	0	0	0	0	
1	0	0	1	0	$F = 0$
1	0	1	0	0	
1	0	1	1	1	$F = D$
1	1	0	0	1	
1	1	0	1	1	$F = 1$
1	1	1	0	1	
1	1	1	1	1	$F = 1$



Realization of Boolean expression using Mux

Realize $F(A, B, C, D) = m_1 + m_3 + m_4 + m_{11} + m_{12} + m_{13} + m_{14} + m_{15}$

<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>F</i>	
0	0	0	0	0	$F = D$
0	0	0	1	1	
0	0	1	0	0	$F = D$
0	0	1	1	1	
0	1	0	0	1	$F = D'$
0	1	0	1	0	
0	1	1	0	0	$F = 0$
0	1	1	1	0	
1	0	0	0	0	$F = 0$
1	0	0	1	0	
1	0	1	0	0	$F = D$
1	0	1	1	1	
1	1	0	0	1	$F = 1$
1	1	0	1	1	
1	1	1	0	1	$F = 1$
1	1	1	1	1	



Use an 8 input MUX to implement the following equation:

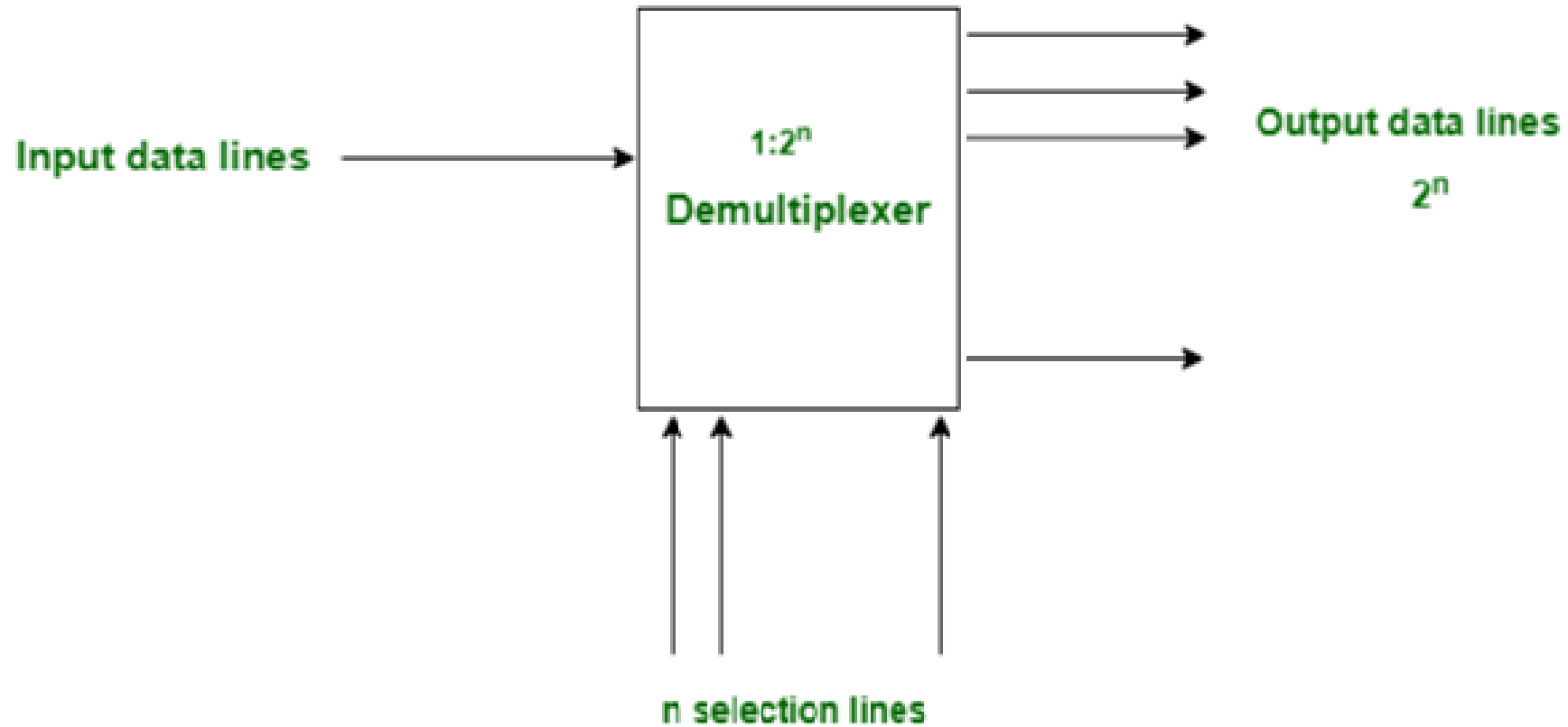
$$Y = \overline{A}.\overline{B}.\overline{C}.\overline{D} + \overline{A}.\overline{B}.C.D + \overline{A}.B.\overline{C}.D + \overline{A}.B.\overline{C}.\overline{D} + A.\overline{B}.\overline{C}.D + A.\overline{B}.C.\overline{D} \\ + A.B.\overline{C}.\overline{D} + A.B.\overline{C}.D$$

$$\overline{A}\overline{B} + ABD + AB\overline{D} + \overline{A}\overline{C}\overline{D} + \overline{A}BC\overline{C}$$

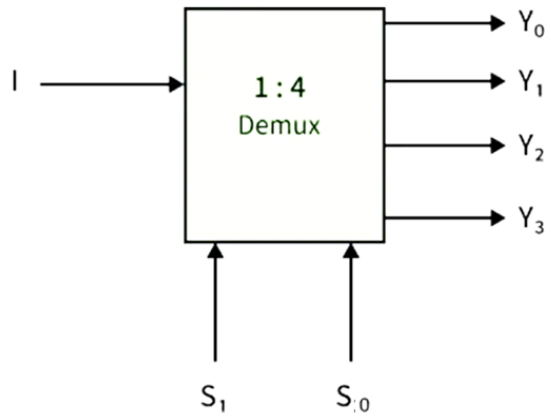
Realization of Boolean expression using Mux

$$\overline{A}\overline{B} + ABD + AB\overline{D} + \overline{A}\overline{C}\overline{D} + \overline{A}B\overline{C}$$

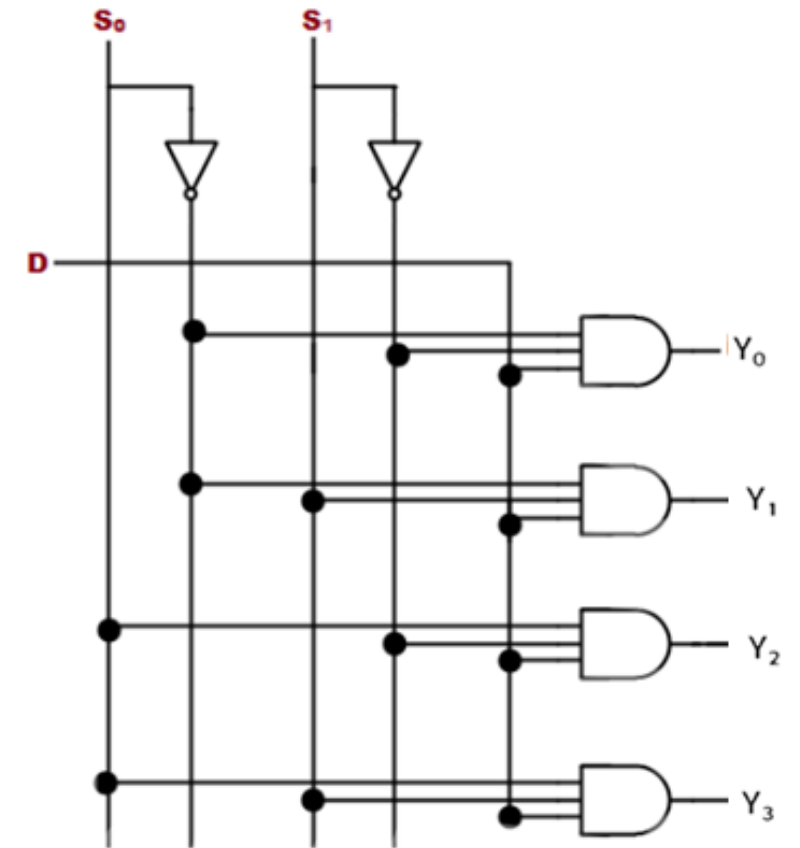
Demultiplexer (DEMUX)



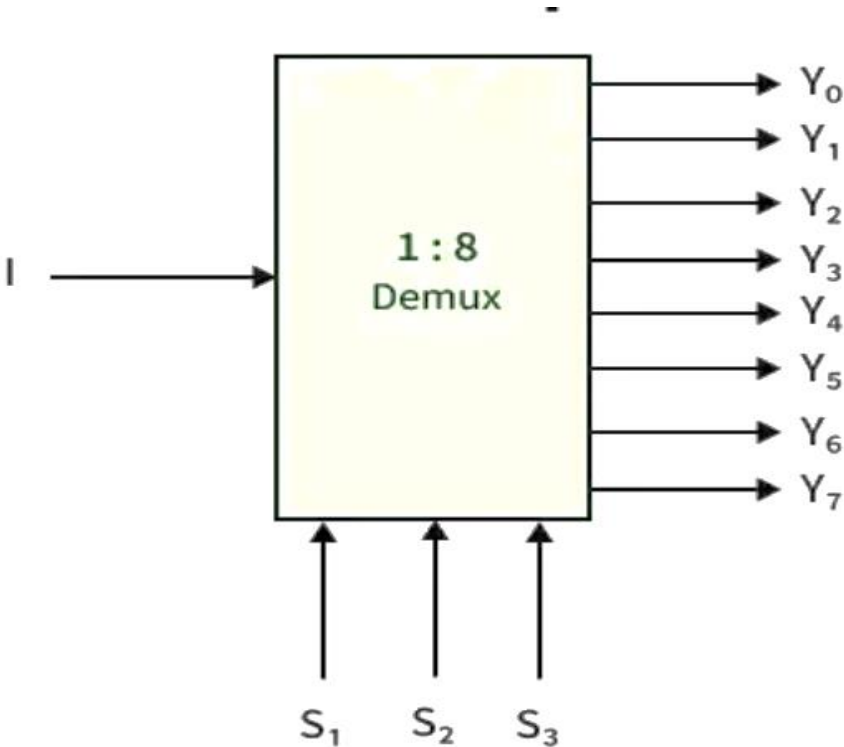
1x4 DEMUX



Selection Inputs		Outputs			
S_1	S_0	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	I
0	1	0	0	I	0
1	0	0	I	0	0
1	1	I	0	0	0

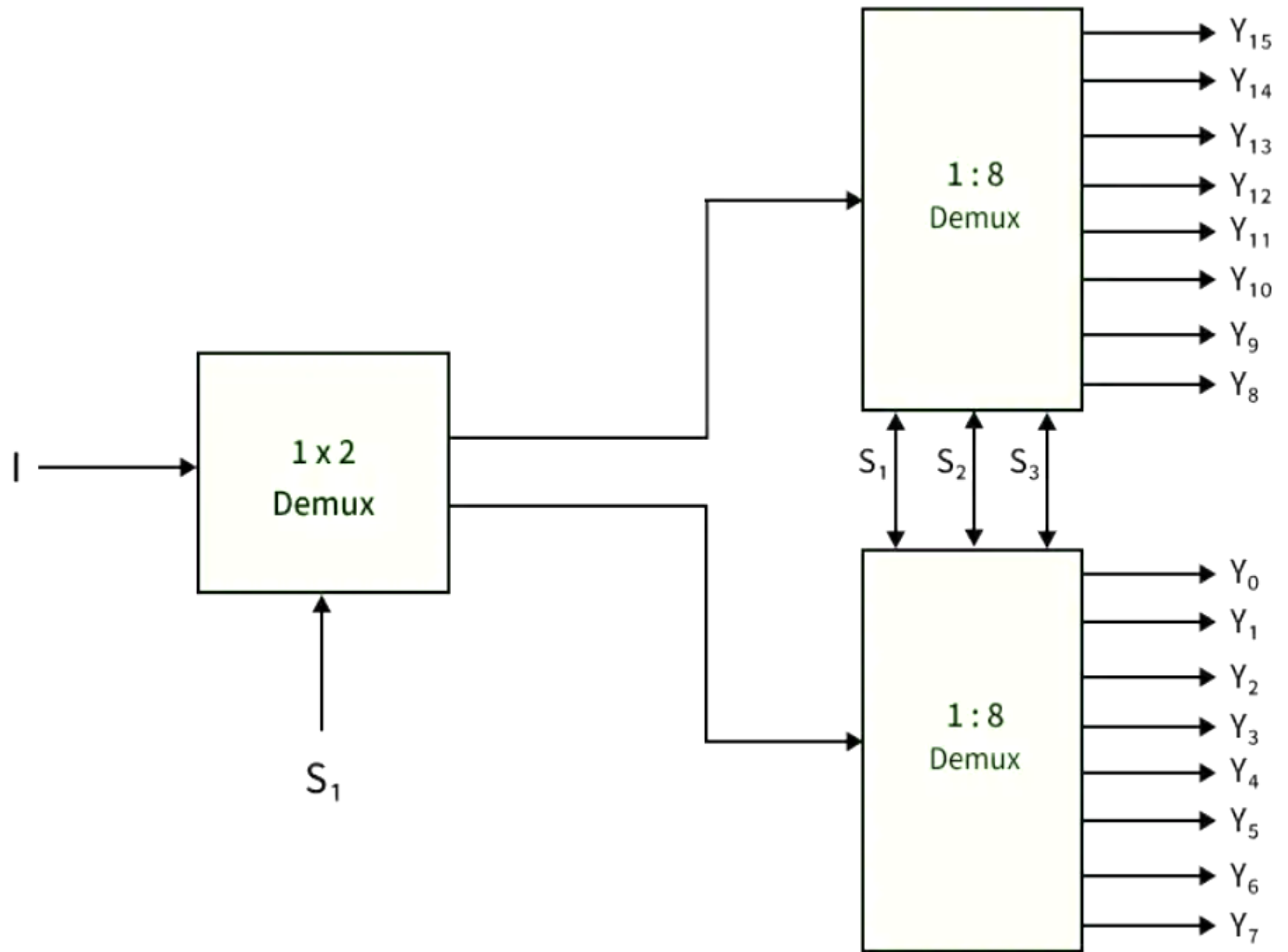


1x8 DEMUX



Selection Inputs			Outputs							
S_2	S_1	S_0	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	1	0	0	0	0	0	0	1	0	0
0	1	1	0	0	0	0	1	0	0	0
1	0	0	0	0	0	1	0	0	0	0
1	0	1	0	0	1	0	0	0	0	0
1	1	0	0	1	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0

1:16 DEMUX
using
1:8 DEMUX



Full Adder using DEMUX

A	B	Cin	Sum (S)	Carry (Cout)
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

